

Basics of Cloud Computing

Lesson 1: Microsoft Azure

Cloud Computing & Big Data Analytics — Graduate Programme in Digital Science | Day 11 | March 2026

Today's Agenda

- What is cloud computing and why it matters
- Microsoft Azure — platform overview and history
- Core service categories for data scientists
- The Shared Responsibility Model
- Azure's global infrastructure (regions, zones)
- Portal, CLI, and resource management interfaces
- Pricing, free tier, and cost control strategies
- Certifications and career paths in Azure

01

Understanding Cloud Computing

The Cloud Computing Analogy

- Think of a small bakery that bakes 50 cupcakes on a regular day
- On Valentine's Day, demand explodes to 5,000 cupcakes
- Option A: Buy 10 giant ovens that sit idle 364 days/year → On-premises IT
- Option B: Rent extra oven space only when you need it → Cloud computing
- Cloud computing = renting IT resources (compute, storage, databases) on demand
- You pay only for what you consume, by the minute

KEY TAKEAWAY: Cloud computing means renting IT resources on demand instead of owning them — you pay only for what you consume.

What Is Microsoft Azure?

- Cloud platform built and operated by Microsoft Corporation
- Launched February 2010 as 'Windows Azure', rebranded 2014
- 2nd largest cloud provider globally (behind AWS)
- Operates across 60+ regions — more than any other cloud provider
- Over 600 services: compute, storage, AI/ML, databases, networking, security
- Powers enterprises from Fortune 500 companies to startups worldwide

KEY TAKEAWAY: Azure is not just server rental — it is an integrated ecosystem of 600+ services managed through a unified control plane.

Azure by the Numbers

600+

Cloud Services

Compute, Storage, AI,
Databases, Networking

60+

Global Regions

More geographic
coverage
than any competitor

95%

Fortune 500

Companies using
Microsoft Azure

\$200

Free Credit

To start exploring
Azure services

Why Azure Matters to Data Scientists

- 1. Microsoft Ecosystem Integration — seamless with Microsoft 365, Power BI, SQL Server, unified authentication
- 2. Managed ML Platform — Jupyter notebooks, auto-scaling GPU clusters, model registry, REST API deployment
- 3. Exclusive OpenAI Partnership — only enterprise cloud host for GPT models, critical for LLM applications
- 4. Comprehensive Analytics — Synapse, Databricks, HDInsight for big data workloads
- 5. Global Compliance — 90+ compliance certifications (GDPR, HIPAA, SOC, ISO)

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Cloud Service Models

Service Models — IaaS vs PaaS vs SaaS

Model	You Manage	Azure Manages	Example
IaaS (Infrastructure as a Service)	OS, apps, data	Hardware, networking	Azure Virtual Machines
PaaS (Platform as a Service)	Apps, data only	OS, runtime, hardware	Azure ML, App Service
SaaS (Software as a Service)	Data only	Everything else	Microsoft 365, Outlook

KEY TAKEAWAY: As you move from IaaS → PaaS → SaaS, you hand more operational burden to Microsoft and keep more time for data science.

The Shared Responsibility Model

Layer	On-Premises	IaaS	PaaS	SaaS
Physical data centre	You	Microsoft	Microsoft	Microsoft
Network controls	You	You	Microsoft	Microsoft
Operating system	You	You	Microsoft	Microsoft
Applications	You	You	You	Microsoft
Data & access	YOU	YOU	YOU	YOU

KEY TAKEAWAY: Your data and who can access it is ALWAYS your responsibility — regardless of service model.

03

Azure Global Infrastructure

Azure's Three-Tier Infrastructure

- REGIONS — Sets of data centres in a geographic area (e.g., Canada Central, West Europe)
- Each region connected via dedicated low-latency backbone network
- AVAILABILITY ZONES — Physically separate locations within a region
- Independent power, cooling, and networking per zone (fault isolation)
- RESOURCE GROUPS — Logical containers for related resources
- Deploy, monitor, bill, and delete resources as a single unit

KEY TAKEAWAY: Deploy close to your users for low latency. Choose regions that comply with data residency laws (GDPR, PIPEDA).

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Core Services for Data Scientists

Core Services – Compute

Service	Model	Data Science Use Case
Azure VM (Virtual Machine)	IaaS	Custom environments, any OS/library
Container Instances (ACI)	Light PaaS	Quick Docker-based model deployment
AKS (Kubernetes Service)	Orchestrated PaaS	Production model serving at scale
Azure Functions	Serverless (FaaS)	Event-triggered data preprocessing
ML Compute	Managed ML	GPU clusters that auto-terminate after training

KEY TAKEAWAY: Choose based on the control-vs-convenience spectrum. VMs give full control; ML Compute minimises overhead.

Core Services — Storage

Storage Types

- Blob Storage — unstructured data (CSVs, images, logs)
- Data Lake Gen2 — hierarchical namespace for Spark/Hadoop
- File Storage — SMB file shares for legacy tools
- Table & Queue — NoSQL key-value and messaging

Storage Tiers (Cost Optimisation)

- Hot — frequent access, highest storage cost
- Cool — infrequent, lower storage, higher retrieval
- Cold — rare access, even cheaper storage
- Archive — cheapest storage, hours to retrieve data

Core Services — AI, ML & Analytics

- Azure Machine Learning — end-to-end lifecycle: AutoML, experiments, model registry, managed endpoints
- Cognitive Services — pre-built AI via API (vision, speech, language, decision)
- Azure OpenAI Service — GPT-4, GPT-4o, DALL-E, Whisper with enterprise security
- Synapse Analytics — unified data warehouse + big data (SQL + Spark in one workspace)
- Azure Databricks — collaborative Spark platform for ETL and large-scale ML training

KEY TAKEAWAY: Azure covers the entire data science workflow — from ingestion to deployment and monitoring.

Core Services — Databases

Database	Type	Best For
Azure SQL Database	Relational (SQL Server)	Structured analytics, experiment metadata
Cosmos DB	NoSQL (multi-model)	Real-time feature stores, sub-10ms lookups
Database for PostgreSQL	Relational (open-source)	Teams already using PostgreSQL
Database for MySQL	Relational (open-source)	Web application backends

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Managing Azure Resources

Five Ways to Manage Azure

- 1. Azure Portal — web-based GUI at portal.azure.com (visual, point-and-click)
 - 2. Azure CLI — cross-platform command line using 'az' commands (scriptable)
 - 3. Azure PowerShell — cmdlets for Windows, macOS, Linux scripting
 - 4. ARM Templates / Bicep — declarative Infrastructure as Code (JSON/Bicep files)
 - 5. SDKs — programmatic libraries for Python, .NET, Java, JavaScript, Go
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- All five interfaces route through Azure Resource Manager (ARM) — consistent results guaranteed

KEY TAKEAWAY: The Portal is great for learning; CLI and Infrastructure as Code are essential for production deployments.

The Azure Portal — Your Command Centre

- Web dashboard at portal.azure.com — works in any modern browser
- Visual resource creation, configuration, and monitoring
- Customisable dashboards: pin charts, resource groups, cost views
- Built-in Cloud Shell — Bash or PowerShell directly in browser
- Role-Based Access Control (RBAC) — assign permissions per team member
- Activity Log — audit trail of every change to every resource

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Pricing & Cost Management

How Azure Charges — Pay-As-You-Go

- Most services billed by the MINUTE — no upfront commitments required
- Scale up (more power) or scale out (more instances) at any time
- Scale to zero when not in use — stop paying when you stop using
- No long-term contracts needed (though they can save money)

KEY TAKEAWAY: Azure's pricing is like a utility bill — pay for what you consume. But unlike electricity, you can turn individual services on and off instantly.

Cost-Saving Strategies

Strategy	Potential Savings	Trade-off
Reserved Instances (1-3 year)	Up to 72%	Upfront commitment required
Spot VMs	Up to 90%	Can be evicted at any time
Auto-scaling to zero	Variable (100% when idle)	Cold start latency on restart
Storage tiering (Cool/Archive)	50-90% on storage cost	Higher retrieval cost & time
Budget alerts (50/80/100%)	Prevention	Requires setup discipline

Azure Free Account — Start Today

\$200

Free Credit

Use on ANY service
for 30 days

12 mo

Free Services

VMs, Blob Storage,
SQL Database (limited)

65+

Always Free

Functions: 1M exec/mo
Cognitive: 5K calls/mo

Essential Cost Management Tools

- Azure Pricing Calculator — estimate monthly costs BEFORE you deploy anything
- Azure Cost Management + Billing — set budget alerts at 50%, 80%, 100%
- Resource Tags — label resources by project, team, or environment for cost tracking
- Azure Advisor — personalised recommendations to reduce cost and improve performance
- Cost Analysis dashboard — drill down into which services cost the most

KEY TAKEAWAY: ALWAYS estimate costs before deploying. Set budget alerts. Delete resources when done. These three habits will save you thousands.

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Certifications & Career Path

Azure Certification Path

Level	Certification	Focus Area	For Data Scientists?
Fundamentals	AZ-900	Cloud concepts, core Azure services	Start here
Associate	DP-100	Data Science Solution on Azure	Core certification
Associate	AI-102	AI Engineering on Azure	If building AI apps
Associate	AZ-104	Azure Administration	If managing infra
Expert	AZ-305	Solutions Architecture	For architects

KEY TAKEAWAY: Start with AZ-900 (fundamentals), then pursue DP-100 (data science). Cloud-certified professionals command higher starting salaries.

DP-100: Data Scientist Associate

- THE certification for Azure data scientists — directly maps to this programme
- Covers Azure ML workspaces, experiments, compute management
- Automated ML (AutoML) — model selection and hyperparameter tuning
- Responsible AI — fairness, interpretability, privacy, transparency
- Model deployment as REST API endpoints with managed monitoring
- Validates real-world skills that employers actively seek

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Azure ML Project Workflow

Typical ML Project on Azure — 6 Steps

- Step 1: INGEST — Upload raw data to Blob Storage or Data Lake Gen2
- Step 2: PREPARE — Clean and transform with Synapse Analytics or Databricks
- Step 3: TRAIN — Build models in Azure ML with auto-scaling GPU compute
- Step 4: EVALUATE — Track experiments, compare metrics across runs
- Step 5: DEPLOY — Serve as REST API via managed online endpoints
- Step 6: MONITOR — Track model drift, performance, costs with Azure Monitor

KEY TAKEAWAY: Azure provides a managed service at every step — from raw data to production prediction endpoint.

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Azure vs the Competition

Azure vs AWS vs GCP — Quick Comparison

Dimension	Azure	AWS	GCP
Market Share (2025)	~24%	~31%	~11%
Global Regions	60+	33	40
ML Platform	Azure ML	SageMaker	Vertex AI
Unique Strength	Microsoft + OpenAI	Broadest catalogue	Data/AI innovation
Free Tier	\$200 / 30 days	\$300 / 12 months	\$300 / 90 days

KEY TAKEAWAY: All three are capable — Azure excels when your organisation uses Microsoft tools or needs OpenAI integration.

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Key Concepts Review

Essential Abbreviations (1/2)

Abbreviation	Full Name	What It Means
VM	Virtual Machine	A software-based computer running on cloud hardware
IaaS	Infrastructure as a Service	You manage OS and apps; cloud manages hardware
PaaS	Platform as a Service	You manage apps and data; cloud manages the rest
SaaS	Software as a Service	You manage data only; everything else is managed
ARM	Azure Resource Manager	The control plane behind all Azure operations
CLI	Command Line Interface	Text-based tool for managing Azure resources

Essential Abbreviations (2/2)

Abbreviation	Full Name	What It Means
API	Application Programming Interface	A way for software to talk to other software
GPU	Graphics Processing Unit	Processor optimised for parallel computation (ML)
ML	Machine Learning	Systems that learn from data without explicit rules
LLM	Large Language Model	AI trained on vast text data (e.g., GPT-4)
RBAC	Role-Based Access Control	Assigning permissions based on team roles
ETL	Extract, Transform, Load	Pipeline to move and reshape data

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Summary & Five Things to Remember

Five Things to Take Away Today

- 1. Azure = 600+ services spanning compute, storage, AI, databases, and networking
- 2. Shared Responsibility — your data security is ALWAYS your responsibility
- 3. Choose the right service model — IaaS for control, PaaS for productivity, SaaS for simplicity
- 4. Cost awareness is mandatory — use Pricing Calculator, set alerts, delete unused resources
- 5. Start free, certify early — Free Account for hands-on learning, AZ-900 → DP-100 for career growth

Recommended Video — Watch Before Next Class

- AZ-900 Episode 1: Cloud Computing and Vocabulary
- By Adam Marczak — Azure for Everyone (Microsoft MVP)
- Duration: approximately 13 minutes
- Link: youtube.com/watch?v=Pt9LeIJ0fL0
- Covers: cloud computing basics, high availability, scalability, elasticity, fault tolerance
- Note: Video used for educational reference only. All IP rights belong to the original creator.

Recommended Resources & Links

- Azure Free Account: azure.microsoft.com/en-us/free/
- Azure Portal: portal.azure.com
- Microsoft Learn (Azure): learn.microsoft.com/en-us/azure/
- Azure Pricing Calculator: azure.microsoft.com/en-us/pricing/calculator/
- AZ-900 Study Guide: marczak.io/az-900/
- Azure Architecture Centre: learn.microsoft.com/en-us/azure/architecture/
- Verify URLs: virustotal.com/gui/home/url

Next Session Preview

- Hands-on Azure environment setup — working with a live subscription
 - Creating real resources: VM, Storage Account, ML Workspace
 - Running your first ML experiment in Azure Machine Learning
 - Exploring the Azure Portal and CLI interactively
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- ACTION ITEM: Create your Azure Free Account before next class!
 - → azure.microsoft.com/en-us/free/

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